

Supplementary Information: Criterion validity of heel-mounted inertial sensors for stride-length estimation during single- and dual-task walking in older adults with and without cognitive impairment

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This document contains the supplementary tables referenced in the main manuscript: per-participant demographics (S1), reported-condition data availability (S2), a MoCA-based cognitive-status sensitivity reanalysis (S3), a same-pair gravity-residual correction ablation (S4), a participant-clustered bootstrap of cohort-level accuracy (S5), a sensitivity analysis of the stride-pair mismatch filter (S6), and a correction-gain sensitivity analysis (S7).

S1. PER-PARTICIPANT DEMOGRAPHICS

Table S1 lists demographic characteristics for all 33 enrolled participants. We excluded one participant (ID.191) because the available recordings used shank and hip sensors rather than the heel-mounted sensors required for this validation. The remaining 32 participants contributed every reported condition for which the paired source data were available.

Table S1. Individual participant demographics. ID = numeric study code. MoCA = Montreal Cognitive Assessment total (/30). Cognitive = MoCA-defined screening group (CN = cognitively normal, MoCA $\geq$ 26; CI = cognitive impairment, MoCA 14–25). These labels do not denote clinical diagnoses. Excluded = $\times$ when the required heel-mounted recordings were unavailable.

ID	Age	Sex	Height (cm)	Mass (kg)	MoCA	Cognitive	Excluded
ID_194	60	F	166.0	89.5	27	CN	
ID_104	80	F	159.0	75.0	17	CI	
ID_171	66	F	168.0	65.4	30	CN	
ID_107	75	F	163.0	66.9	27	CN	
ID_196	71	F	168.4	70.5	28	CN	
ID_139	73	F	165.0	99.8	28	CN	
ID_123	66	M	177.0	77.6	25	CI	
ID_174	68	M	177.0	80.5	23	CI	
ID_183	60	F	158.5	67.5	29	CN	
ID_177	74	M	168.0	78.6	20	CI	
ID_110	67	F	163.0	62.6	30	CN	
ID_180	78	M	181.5	96.5	25	CI	
ID_185	67	F	170.0	61.6	26	CN	
ID_192	60	F	162.0	72.7	26	CN	
ID_188	80	F	160.5	53.1	21	CI	
ID_197	63	F	171.0	72.0	27	CN	
ID_189	63	F	167.5	83.0	24	CI	
ID_191	60	F	165.0	74.2	29	CN	$\times$
ID_190	75	F	158.0	66.7	26	CN	
ID_182	60	F	173.0	81.7	28	CN	
ID_169	71	M	173.0	81.3	27	CN	
ID_186	66	F	165.0	61.3	23	CI	
ID_109	82	F	166.0	75.2	28	CN	
ID_101	83	F	158.0	69.6	28	CN	
ID_184	70	F	170.0	77.5	25	CI	
ID_126	83	F	161.0	50.0	29	CN	
ID_149	69	M	173.0	68.4	22	CI	
ID_114	82	F	163.0	50.4	26	CN	
ID_135	74	F	160.0	74.1	29	CN	
ID_193	75	M	175.0	79.8	24	CI	
ID_195	61	M	192.0	104.5	23	CI	
ID_136	82	F	161.0	67.4	28	CN	
ID_172	61	F	164.0	53.9	23	CI	

S2. REPORTED-CONDITION DATA AVAILABILITY

We retained every reported participant-condition record that had the required source data and produced stride pairs under the common matching rules. We did not remove records based on match count or stride-length error because both are study outcomes. Of the 128 possible records from 32 participants across four conditions, 126 entered the paired analysis. The GR Pulse source record was unavailable for ID_186 L2 and ID_195 L2, so those two conditions could not be aligned to the GaitRite timeline. Table S2 lists these unavailable records.

The 126 available records contained 1,764 walks. We accepted a walk-level alignment when at least 60% of the expected GaitRite heel strikes had an IMU match within 150 ms. A total of 1,669 walks passed this check and 95 were classified as synchronisation failures. These included 14 single-task, 13 serial-2, 19 serial-3, and 49 serial-7 walks. Failed alignments did not enter paired stride or event-agreement analyses. We did not classify them as sensor failures or remove any walk solely because of its stride-length error.

35 Three aligned walks did not enter stride pairing because a valid stride-level comparison was not possible.
 36 ID_107 single-task trial 2 OUT and ID_149 serial-2 trial 1 BACK contained an IMU event before mat
 37 entry that did not correspond to a GaitRite stride. ID_195 serial-7 trial 1 BACK contained inconsistent
 38 GaitRite heel-strike and stride-time records. These walk-level decisions were defined in the shared
 39 analysis configuration and did not remove the remaining walks from those participant-condition records.

Table S2. Reported-condition records unavailable for paired ZUPT-GaitRite analysis. L2 = serial subtraction by 2.

Subject	Condition	Diagnostic note
ID_186	L2	GR Pulse source record unavailable
ID_195	L2	GR Pulse source record unavailable

40 S3. MOCA-BASED COGNITIVE-STATUS SENSITIVITY REANALYSIS

41 **We re-ran the cognitive-status stratification after excluding the one MoCA-defined CI participant**
 42 **whose score fell at the moderate-impairment boundary.** All 32 analysed participants had recorded
 43 MoCA scores (Table S1). The MoCA-defined CI group ($n = 13$) spanned MoCA 17–25. Twelve
 44 participants scored 18–25 and one scored 17, at the boundary of moderate impairment. Restricting the
 45 group to the twelve participants with MoCA scores of 18–25 returned an RMSE of 4.01 cm, compared
 46 with 4.14 cm in the full CI group (Table S3). This result shows that the group estimate does not depend
 47 on the single moderate-range participant.

Table S3. Cognitive-status sensitivity reanalysis. Primary = paper-reported MoCA-defined screening groups (all 32 participants, complete MoCA). Restricted CI = MoCA-defined CI group restricted to the twelve participants with MoCA 18–25, excluding the one participant at MoCA 17. MAE statistics are per-stride pooled.

Rule	Group	n subj	N pairs	RMSE (cm)	MAE (cm)	Bias (cm)
Primary (paper)	CN	19	6,139	3.65	2.29	+0.28
	CI	13	4,333	4.14	2.79	+0.24
Restricted CI (sensitivity)	CN	19	6,139	3.65	2.29	+0.28
	CI (MoCA 18–25)	12	3,963	4.01	2.71	+0.32

48 S4. GRAVITY-RESIDUAL CORRECTION ABLATION

49 **We re-ran the full pipeline on the same 32-participant cohort with the retroactive gravity-residual**
 50 **correction disabled.** The ablation kept stance detection, bias estimation, orientation tracking, and the
 51 per-stride zero-velocity reset unchanged. We removed only the swing-phase gravity-residual ramp. Both

analyses used the same 10,472 paired strides. Disabling the correction increased pooled RMSE by 0.99 cm and shifted bias from +0.3 to +2.2 cm. The proportional-bias slope changed from 0.036 to 0.044, and ICC fell from 0.976 to 0.962 (Table S4).

Table S4. Gravity-residual correction ablation on the same 10,472 paired strides. ON = canonical pipeline and OFF = ablation. Bias = IMU minus GaitRite. Slope = proportional-bias regression of stride-level error on the mean of paired stride lengths.

Condition	Correction	RMSE (cm)	MAE (cm)	Bias (cm)	ICC(2,1)	Slope
Pooled	OFF	4.85	3.42	+2.25	0.962	0.044
	ON	3.86	2.50	+0.26	0.976	0.036
Single-task	OFF	5.25	3.98	+2.94	0.955	0.052
	ON	3.96	2.72	+0.71	0.973	0.033
Serial 2s	OFF	4.49	3.10	+1.84	0.966	0.029
	ON	3.70	2.39	+0.07	0.977	0.033
Serial 3s	OFF	4.78	3.23	+1.98	0.963	0.024
	ON	3.91	2.41	+0.03	0.975	0.026
Serial 7s	OFF	4.87	3.39	+2.26	0.956	0.066
	ON	3.87	2.49	+0.26	0.971	0.048

S5. BOOTSTRAP ROBUSTNESS OF COHORT-LEVEL ACCURACY

A participant-clustered bootstrap shows stable cohort-level accuracy under participant resampling. We resampled the 32 analysed participants with replacement 10,000 times and recomputed each accuracy metric on the resulting concatenated stride pool. Because participants contributed different numbers of strides, the pool size varied across resamples. The 95% percentile confidence intervals are shown in Table S5.

Table S5. Cohort-level accuracy metrics with 95% confidence intervals.

Metric	Point estimate	95% CI	CI half-width
RMSE (cm)	3.86	[3.51, 4.23]	0.36
MAE (cm)	2.50	[2.30, 2.70]	0.20
Bias (cm)	+0.26	[-0.08, +0.63]	0.35
ICC(2,1)	0.976	[0.964, 0.981]	0.008

61 S6. SENSITIVITY TO THE STRIDE-PAIR MISMATCH FILTER

62 **Removing the stride-pair mismatch filter retained 22 additional strides and left typical absolute**
63 **error similar.** This filter required IMU stride length and stride time to remain within 50–150% of their
64 matched GaitRite values. We retained all other participant, condition, walk, GaitRite plausibility, on-mat,
65 and mutual-nearest matching rules. The resulting pool contained 10,494 rather than 10,472 strides. Pooled
66 RMSE rose from 3.86 to 5.49 cm, while MAE changed from 2.50 to 2.65 cm (Table S6).

67 The filter removed 22 strides (0.21% of the unfiltered pool). Their median stride-length error was
68 –72.2 cm, RMSE was 85.2 cm, and maximum absolute error was 129.2 cm. These values are consistent
69 with missed or merged stride events or unstable stride estimates rather than ordinary estimator error. The
70 difference between pooled RMSE and MAE therefore reflects RMSE’s sensitivity to a small number of
71 large mismatches rather than a broad shift in typical stride accuracy.

Table S6. Stride-length accuracy with and without the stride-pair mismatch filter. The filter restricted IMU stride length and stride time to 50–150% of the matched GaitRite value. No mismatch filter = both checks omitted, with all other filtering and matching rules unchanged. Bias = IMU minus GaitRite.

Analysis	<i>N</i> pairs	RMSE (cm)	MAE (cm)	Bias (cm)	ICC(2,1)
Mismatch filter	10,472	3.86	2.50	+0.26	0.976
No mismatch filter	10,494	5.49	2.65	+0.17	0.952

72 S7. GRAVITY-RESIDUAL CORRECTION GAIN SENSITIVITY

73 **Stride-length accuracy was stable across retroactive gravity-residual correction gains from 0.3 to 0.7.**
74 We varied only the correction gain from 0.3 to 0.7 in increments of 0.1. All settings used the canonical
75 stance thresholds, filtering, matching, and walk-alignment rules. RMSE changed smoothly from 4.10 cm
76 at 0.3 to 3.83 cm at 0.6 and 3.87 cm at 0.7 (Table S7). The canonical gain of 0.5 produced 3.86 cm RMSE,
77 within 0.03 cm of the smallest value. The selected gain therefore lies in a broad near-optimal region
78 rather than at a sharp optimum. Because all gains were evaluated in the same cohort, this analysis tests
79 parameter stability but does not replace independent validation.

Table S7. Sensitivity of pooled stride-length accuracy to the retroactive gravity-residual correction gain.
Bias = IMU minus GaitRite.

Gain	<i>N</i> pairs	RMSE (cm)	MAE (cm)	Bias (cm)	ICC(2,1)
0.3	10,472	4.10	2.71	+1.05	0.973
0.4	10,472	3.95	2.57	+0.66	0.975
0.5 (canonical)	10,472	3.86	2.50	+0.26	0.976
0.6	10,472	3.83	2.49	−0.13	0.976
0.7	10,472	3.87	2.54	−0.52	0.976